NETWORKING / NET01 / OCTOBER 2026

Segmented Office Network with Cisco Switching & Routing

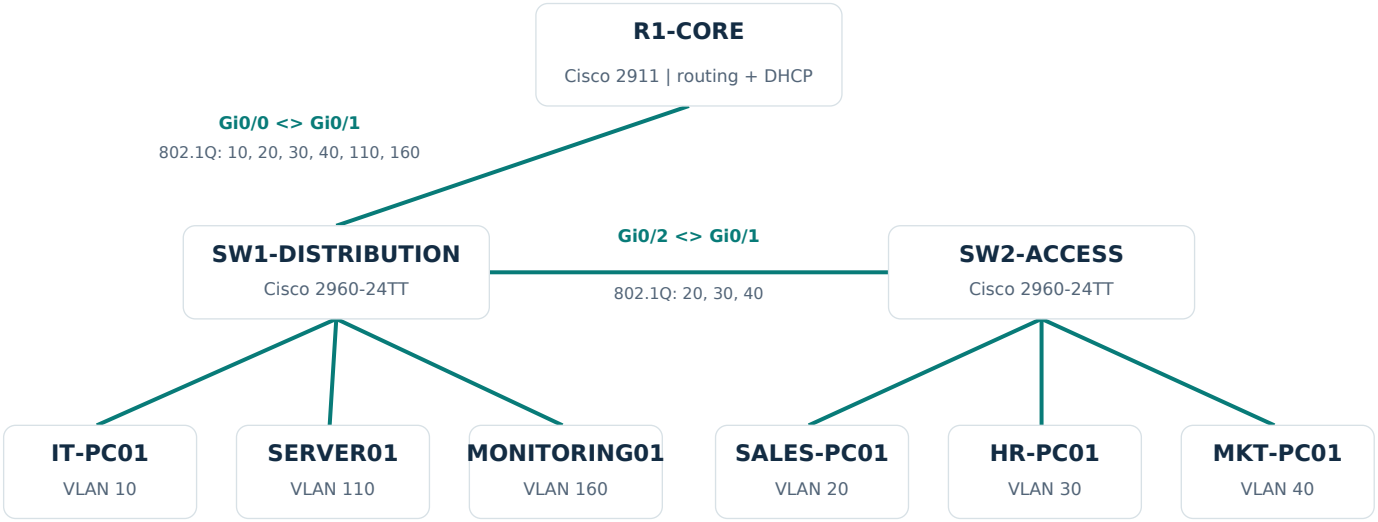
A Packet Tracer extension of INFRA-BASE-v1

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The objective

I rebuilt the networking side of my Balcom Insurance homelab in Cisco Packet Tracer. The goal was to practice the switch and router work behind a segmented office: assign endpoint ports, carry VLANs over trunks, route between subnets, and provide reliable client addressing.

Balcom Insurance is fictional. This project is a simulation in my personal lab, not a production deployment or client engagement.



Logical topology redrawn from the NET01 build. One building; no WAN or redundant links.

6 VLANs	2 switches	4 DHCP clients	2 static servers
Separate subnets	Cisco 2960-24TT	Department workstations	Servers + Monitoring

Result

All six router subinterfaces reached up/up. The router issued a lease in each departmental subnet, and the captured gateway and cross-subnet ICMP tests completed without packet loss. The completed build provides a baseline for deliberate troubleshooting and later network expansion.

01 / ARCHITECTURE

Keep the addressing familiar; make switching explicit

I kept the department and infrastructure subnets from the source architecture. In the original lab, OPNsense routed the networks. Here, a Cisco 2911 performs that role through 802.1Q subinterfaces on one physical link.

VLAN	Purpose	Network /24	Gateway	Endpoint observed
10	IT	10.20.10.0	10.20.10.1	10.20.10.100
20	SALES	10.20.20.0	10.20.20.1	10.20.20.100
30	HR	10.20.30.0	10.20.30.1	10.20.30.100
40	MARKETING	10.20.40.0	10.20.40.1	10.20.40.100
110	SERVERS	10.10.20.0	10.10.20.1	10.10.20.100
160	MONITORING	10.10.30.0	10.10.30.1	10.10.30.100

The Servers and Monitoring networks were separate interfaces in the source design. VLAN 110 and VLAN 160 are new mappings for this Packet Tracer adaptation. The unused Clients network, isolated malware network, and home/WAN connection are outside this build.

Physical port map

Device / port	Connected device / port	Role
R1 Gi0/0	SW1 Gi0/1	Trunk: all six VLANs
SW1 Gi0/2	SW2 Gi0/1	Trunk: VLANs 20, 30, 40
SW1 Fa0/1	IT-PC01 FastEthernet0	Access VLAN 10
SW1 Fa0/2	SERVER01 FastEthernet0	Access VLAN 110
SW1 Fa0/3	MONITORING01 FastEthernet0	Access VLAN 160
SW2 Fa0/1	SALES-PC01 FastEthernet0	Access VLAN 20
SW2 Fa0/2	HR-PC01 FastEthernet0	Access VLAN 30
SW2 Fa0/3	MKT-PC01 FastEthernet0	Access VLAN 40

The two 2960 switches provide Layer 2 connectivity. The distribution label describes SW1's position in this small design; inter-VLAN routing remains on R1.

02 / SWITCHING

Access ports and trunks serve different purposes

I created the six VLANs on both switches and assigned endpoint ports explicitly as access ports. The membership tables show IT, Servers, and Monitoring on SW1; Sales, HR, and Marketing on SW2.

```
SW1-DISTRIBUTION#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   IT                               active    Fa0/1
20   SALES                            active
30   HR                               active
40   MARKETING                        active
110  SERVERS                          active    Fa0/2
160  MONITORING                       active    Fa0/3
1002 fddi-default                     active
1003 token-ring-default               active
```

Figure 1. SW1: Fa0/1 in VLAN 10, Fa0/2 in VLAN 110, and Fa0/3 in VLAN 160.

```
SW2-ACCESS#show vlan br
SW2-ACCESS#show vlan brief

VLAN Name                             Status    Ports
---- -------------------------------- --------- -------------------------------
1    default                          active    Fa0/4, Fa0/5, Fa0/6, Fa0/7
                                                Fa0/8, Fa0/9, Fa0/10, Fa0/11
                                                Fa0/12, Fa0/13, Fa0/14, Fa0/15
                                                Fa0/16, Fa0/17, Fa0/18, Fa0/19
                                                Fa0/20, Fa0/21, Fa0/22, Fa0/23
                                                Fa0/24, Gig0/1, Gig0/2
10   IT                               active
20   SALES                            active    Fa0/1
30   HR                               active    Fa0/2
40   MARKETING                        active    Fa0/3
110  SERVERS                          active
160  MONITORING                       active
1002 fddi-default                     active
1003 token-ring-default               active
1004 fddinet-default                  active
1005 trnet-default                    active
SW2-ACCESS#
```

Figure 2. SW2: Fa0/1 in VLAN 20, Fa0/2 in VLAN 30, and Fa0/3 in VLAN 40.

The inter-switch trunk allows only VLANs 20, 30, and 40 because those are the endpoint VLANs needed on SW2. The router-facing trunk carries all six. Both ends retain native VLAN 1. A VLAN existing in the switch database does not automatically mean it is allowed across every trunk.

03 / ROUTING & TRUNK VALIDATION

One physical router link, six gateways

Each R1 subinterface uses an 802.1Q VLAN tag and the .1 gateway address for its subnet. The parent Gi0/0 interface has no IP address and is enabled with no shutdown.

```
R1-CORE#show ip interra
R1-CORE#show ip interface brief
Interface              IP-Address      OK? Method Status                Protocol
GigabitEthernet0/0     unassigned      YES unset  up                    up
GigabitEthernet0/0.10  10.20.10.1      YES manual up                    up
GigabitEthernet0/0.20  10.20.20.1      YES manual up                    up
GigabitEthernet0/0.30  10.20.30.1      YES manual up                    up
GigabitEthernet0/0.40  10.20.40.1      YES manual up                    up
GigabitEthernet0/0.110 10.10.20.1      YES manual up                    up
GigabitEthernet0/0.160 10.10.30.1      YES manual up                    up
GigabitEthernet0/1     unassigned      YES unset  administratively down down
GigabitEthernet0/2     unassigned      YES unset  administratively down down
Vlan1                  unassigned      YES unset  administratively down down
R1-CORE#
```

Figure 3. R1 Gi0/0 and all six configured subinterfaces are up/up. Unused interfaces remain administratively down.

Example: the IT gateway

```
interface GigabitEthernet0/0.10
 description IT_GATEWAY
 encapsulation dot1Q 10
 ip address 10.20.10.1 255.255.255.0
```

The interface command selects the logical interface; description labels its role. encapsulation dot1Q 10 associates tagged VLAN 10 traffic with it. ip address supplies the gateway. The .10 suffix is a naming convention; the encapsulation command determines the actual tag.

```
SW2-ACCESS#show interfaces trunk
Port        Mode         Encapsulation  Status        Native vlan
Gig0/1      on           802.1q         trunking      1

Port        Vlans allowed on trunk
Gig0/1      20,30,40

Port        Vlans allowed and active in management domain
Gig0/1      20,30,40

Port        Vlans in spanning tree forwarding state and not pruned
Gig0/1      20,30,40
```

Figure 4. The final SW2 trunk output shows VLANs 20, 30, and 40 allowed, active, and forwarding after spanning-tree convergence.

A Sales-to-Server packet travels from SW2 VLAN 20 over the trunk to SW1, then to R1 for routing. It returns to SW1 tagged for VLAN 110 and exits the server access port.

04 / ADDRESS MANAGEMENT

DHCP for workstations; static addresses for servers

I first tested the endpoints with static addressing, then moved the four workstations to DHCP. R1 hosts one pool per department and excludes .1 through .99 in each pool. The gateway stays at .1, and the first observed workstation lease in each subnet is .100.

```
R1-CORE#show ip dhcp binding
IP address       Client-ID/              Lease expiration        Type
                 Hardware address
10.20.10.100     00D0.582A.AC68          --                      Automatic
10.20.20.100     00E0.A347.D055          --                      Automatic
10.20.30.100     00D0.BAEB.C618          --                      Automatic
10.20.40.100     0007.EC4E.04DB          --                      Automatic
R1-CORE#
```

Figure 5. Four automatic bindings: IT, Sales, HR, and Marketing.

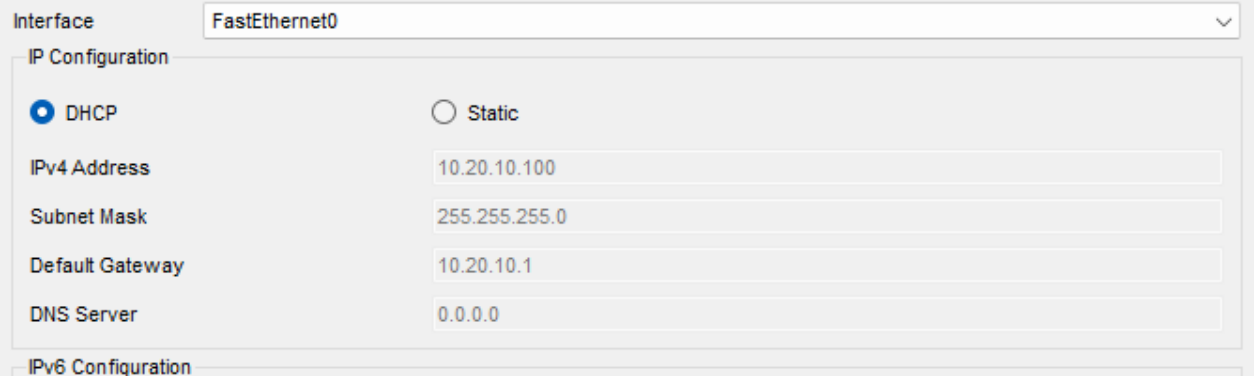


Figure 6. IT-PC01 has DHCP selected and received 10.20.10.100/24 with gateway 10.20.10.1. DNS remains unset.

Example: the IT pool

```
ip dhcp excluded-address 10.20.10.1 10.20.10.99
ip dhcp pool IT
 network 10.20.10.0 255.255.255.0
 default-router 10.20.10.1
```

excluded-address protects the reserved range. pool names the scope; network defines its subnet; default-router supplies the client gateway. Separate pools use the same pattern for VLANs 20, 30, and 40.

No DHCP relay is needed here because R1 directly serves the client VLANs through its own subinterfaces. SERVER01 stays at 10.10.20.100; MONITORING01 stays at 10.10.30.100. Neither infrastructure subnet has a DHCP pool in NET01.

05 / CLIENT VALIDATION

Check the result from the users' side

After the four DHCP bindings appeared, I verified access to SERVER01 from Sales, HR, and Marketing. Each captured test received all four ICMP replies with 0% loss.

```
C:\>ping 10.10.20.100

Pinging 10.10.20.100 with 32 bytes of data:

Reply from 10.10.20.100: bytes=32 time=1ms TTL=127
Reply from 10.10.20.100: bytes=32 time=1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 7. Sales to SERVER01 (10.10.20.100): 4 sent, 4 received.

```
C:\>ping 10.10.20.100

Pinging 10.10.20.100 with 32 bytes of data:

Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 8. HR to SERVER01: 4 sent, 4 received.

```
C:\>ping 10.10.20.100

Pinging 10.10.20.100 with 32 bytes of data:

Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 9. Marketing to SERVER01: 4 sent, 4 received.

Earlier screenshots also confirmed each department's gateway, HR to IT, Marketing to HR, and IT to SERVER01 after DHCP. These are specific reachability tests; they are not an exhaustive all-pairs or application-service test.

06 / INFRASTRUCTURE VALIDATION

Verify the server networks in both directions

I tested the static infrastructure endpoints separately. Each server reached its own gateway, and Servers-to-Monitoring connectivity succeeded in both directions.

```
C:\>ping 10.10.20.1

Pinging 10.10.20.1 with 32 bytes of data:

Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255
Reply from 10.10.20.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.20.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.10.30.100

Pinging 10.10.30.100 with 32 bytes of data:

Reply from 10.10.30.100: bytes=32 time<1ms TTL=127
Reply from 10.10.30.100: bytes=32 time<1ms TTL=127
Reply from 10.10.30.100: bytes=32 time<1ms TTL=127
Reply from 10.10.30.100: bytes=32 time<1ms TTL=127

Ping statistics for 10.10.30.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>
```

Figure 10. SERVER01 reaches 10.10.20.1 and MONITORING01 at 10.10.30.100 with 0% loss.

```
C:\>ping 10.10.30.1

Pinging 10.10.30.1 with 32 bytes of data:

Reply from 10.10.30.1: bytes=32 time<1ms TTL=255
Reply from 10.10.30.1: bytes=32 time<1ms TTL=255
Reply from 10.10.30.1: bytes=32 time=1ms TTL=255
Reply from 10.10.30.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.10.30.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>ping 10.10.20.100

Pinging 10.10.20.100 with 32 bytes of data:

Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time<1ms TTL=127
Reply from 10.10.20.100: bytes=32 time=1ms TTL=127

Ping statistics for 10.10.20.100:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
Approximate round trip times in milli-seconds:
    Minimum = 0ms, Maximum = 1ms, Average = 0ms

C:\>
```

Figure 11. MONITORING01 reaches 10.10.30.1 and SERVER01 at 10.10.20.100 with 0% loss.

A separate SERVER01-to-Sales test also succeeded, exercising the inter-switch path from the infrastructure side. These tests validate simulated IP reachability; the Server-PT nodes do not reproduce Active Directory, SMB permissions, or Wazuh.

07 / TROUBLESHOOTING

Read the state before changing the configuration

Observation	What I checked / learned
Temporary spanning-tree warnings	While only one side of the switch link was configured as a trunk, the other side reported an 802.1Q BPDU on a non-trunk port. I completed the matching trunk configuration, allowed convergence, then checked the final state.
Initially only VLAN 40 forwarding	The initial output was not the settled result. A later show interfaces trunk listed 20, 30, and 40 forwarding. Per-VLAN spanning-tree checks showed forwarding port states.
Router trunk operationally down	SW1 was configured as a trunk before R1 Gi0/0 was enabled. Enabling the router interface brought the link up; the subinterfaces then showed up/up.
A show command failed in config mode	I returned to privileged EXEC with end before running show ip interface brief. The prompt identifies the command context.
Only one DHCP binding appeared	IT had requested a lease; the other PCs still needed DHCP selected. Once they requested addresses, four bindings appeared. A configured pool and an issued lease are different things.
Ping to 10.10.20.11 failed	The intended server is 10.10.20.100. The corrected target replied. I checked the destination before treating the result as a routing fault.

A repeatable diagnostic sequence

Start with the affected endpoint: address, mask, gateway, then gateway reachability. Check the access-port VLAN and link state. Follow the path through trunk allowed lists and spanning-tree forwarding, then inspect the router subinterface. Compare with a working department to narrow the fault.

Why it matters for network support

The work translates directly to first-line LAN troubleshooting, hardware port mapping, IP inventory, and clear escalation notes. It also strengthens the networking foundation behind my security projects: understanding where a flow should travel makes later monitoring and access-control work more reliable.

08 / OUTCOME, LIMITS & HANDOFF

A working baseline for the next networking labs

What NET01 demonstrates

I configured six VLANs on two Cisco switches, restricted trunk VLAN lists to the required paths, built six router-on-a-stick gateways, and created four DHCP pools. I verified interface state, lease issuance, gateway reachability, and selected cross-subnet paths from both workstation and server networks.

What this project does not claim

Boundary	Current scope
Security policy	VLANs separate Layer 2 broadcast domains, but R1 currently routes between the connected subnets. No ACLs or firewall rules restrict that traffic.
Services and external access	DNS is unset. There is no Internet/WAN, NAT, real AD, file service, or Wazuh deployment in Packet Tracer.
Resilience and performance	No redundant paths, EtherChannel, dynamic routing, load testing, or failover testing was performed.
Platform experience	This is simulated Cisco IOS practice. It is not a production deployment, an Arista migration, or validation of enterprise hardware features.
Persistence evidence	Saving device configs and the .pkt file was the closing instruction. No saved-config or reboot evidence was supplied, so persistence is not claimed as tested.

Reproduction and project files

The companion configuration-reference.md includes reconstructed switch and router commands, the DHCP pools, endpoint settings, and verification commands. It is a reference built from the documented session, not an exported running configuration. A .pkt file was not supplied for this publication.

Next in the NET series

NET02 will introduce controlled faults into a separate copy of the baseline. Later work will cover switch replacement and rollback, a second building, routing between sites, redundancy, and access controls. Those are planned extensions, not completed NET01 results.

Portfolio: carlostibj.github.io/project-net01.html
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